

UTKARSH ROY

Technical Assistant

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🌐 LinkedIn Profile 📁 Portfolio Profile

PROFESSIONAL SUMMARY

Electrical Engineer and Technical Assistant with over six years of experience managing undergraduate teaching laboratories and supporting postgraduate research at IIT Delhi. Responsible for laboratory operations serving approximately 400 undergraduate and 75 postgraduate students annually, with expertise in embedded systems, hardware prototyping, laboratory infrastructure, and technical training. Currently pursuing a PhD in Electrical Engineering focused on Battery Management Systems, Embedded Systems, and Edge AI.

TECHNICAL SKILLS

- **Hardware Worked With:**

- **Microcontrollers:** Raspberry Pi Foundation (RP2040, PICO, PICO W, PICO 2W), Texas Instruments (TIVA C, MSP 430), Nordic Semiconductors (nRF52 Series), STM32 Boards, Atmel Boards, Espressif Boards (ESP32, ESP8266), Cypress Semiconductors (PSOC 5, PSOC 6)

- **FPGA & CPLD:** Xilinx/AMD (Spartan 3, Spartan 3E, Zynq 7000 Series, Nexys), Intel Cyclone IV, Renesas (ForgeFPGA), Intel Max II

- **SBC:** R-Pi 3, 4, 5, NVIDIA Jetson Orin, Beaglebone Black

- **Hardware Programming IDEs:** Arduino IDE, Platform IO (in VS Code), Thonny, STM32CubeIDE, STM32CubeMX, PSOC Creator, AMD Vivado Design Suite, Xilinx ISE, Quartus Prime Design Software, Go Configure Software Hub, Code Composer Studio, Energia IDE

- **Software:** C/C++, Python, HTML, Git, Docker, Windows, Linux (Debian, Arch, Fedora), Edge AI, and TinyML applications

- **Design and Fabrication:** LTspice, MATLAB, KiCAD, EasyEDA, FreeCAD, RDWorks

- **Noting and Drafting:** $\LaTeX$, Markdown, AsciiDoc

PROFESSIONAL EXPERIENCE

Technical Assistant

May, 2020 – Present

Dept. of Electrical Engineering, Indian Institute of Technology Delhi, New Delhi, India

- Managed the day-to-day operations and maintenance of the undergraduate Electrical Engineering laboratory, supporting practical coursework for approximately 350 third-year undergraduate students annually while ensuring safe, reliable, and uninterrupted laboratory operations.
- Procured scientific equipment and laboratory accessories by identifying technical requirements and coordinating purchases, resulting in optimised lab resources and updated experimental setups.
- Provided technical instruction and laboratory support for third-year undergraduate students each academic year, assisting with experiments, hardware debugging, and project development.
- Assisted postgraduate students with research experiments by providing technical guidance and troubleshooting, ensuring the seamless execution of graduate-level academic work.
- Administered the laboratory server, performing routine maintenance and system updates, ensuring secure, reliable, and uninterrupted network access for all lab users.
- Delivered hands-on training in laser cutting and 3D printing to undergraduate and postgraduate students, promoting safe fabrication practices and rapid prototyping.
- Mentored undergraduate design projects and postgraduate research by providing technical consultation on embedded hardware, PCB design, fabrication, and system integration.

Graduate Engineer Trainee, Electrical

June, 2018 – August 2018

EMC Limited, Kolkata, India

- Executed the construction of T1 and T2 bus bar towers and supported HT-LT transformation processes, resulting in a robust infrastructure for the electrification project.
- Assisted in the laying of catenary and contact wires, including dropper installation, resulting in reliable overhead line alignment.
- Conducted breakdown tests on transformer oil and facilitated the oil filtration process for a new traction substation transformer, resulting in verified operational safety.

PROJECTS

Edge AI-Based Image Classification Using FOMO on ESP32 Fish-eye Camera

- Developed a real-time image classification system using the Edge Impulse FOMO (Faster Objects, More Objects) framework on an ESP32 Fish-eye Camera.
- Collected and annotated a custom image dataset, trained and optimised the embedded vision model for deployment on a resource-constrained microcontroller.
- Evaluated on-device inference performance, memory utilisation, and classification accuracy for Edge AI applications.

Machine Learning-Based State of Charge Estimation for Lithium-Ion Batteries

- Developed regression-based machine learning models for estimating the State of Charge (SoC) of lithium-ion batteries using experimentally acquired charge-discharge datasets.
- Performed battery data acquisition, pre-processing, feature engineering, model training, and comparative evaluation of prediction performance.
- Investigated the applicability of data-driven SoC estimation techniques for Battery Management System (BMS) applications.

RESEARCH INTERESTS

Battery Management System(BMS), Estimation Algorithms of LIB, Inter-device Communication, Optical-based Free Space Communication, Wearable and Embedded Systems, Edge AI/Tiny ML, Internet of Things.

CERTIFICATIONS

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| • Two Week Faculty Development Program on "AI and EV Synergy" , NIT Patna | Jul. 2025 |
| • Bluetooth Low Energy Fundamentals , Nordic Semiconductor | Sept. 2024 |
| • nRF Connect SDK Fundamentals , Nordic Semiconductor | Sept. 2024 |
| • IEEE Certificates Program (CEU/PDH): 15 Professional Development Hours , IEEE Educational Activities | Feb. 2023 |
| • IEEE Certificates Program (CEU/PDH): 15 Professional Development Hours , IEEE Educational Activities | Aug. 2022 |

EDUCATION

Doctor of Philosophy in Electrical Engineering <i>Indian Institute of Technology Delhi</i> , [New Delhi, India]	July, 2023 - Present
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Relevant Coursework: Embedded Systems and Applications, Introduction to Electric Vehicles, Operations Research, Sensors and Transducers, Biomedical Sensing, Laser-Based Instruments, Batteries for Electric Vehicles, Charging Infrastructure for Electric Vehicles, Application of Machine Learning in EV Diagnostics and Prognostics.

Bachelor of Technology in Electrical Engineering <i>Bengal College of Engineering and Technology, MAKAUT</i> , Durgapur (WB), India DGPA: 8.98/10.0	July, 2018
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Intermediate <i>Gyan Niketan, CBSE</i> , Patna (BR), India Percentage: 90.8%	April, 2014
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Matriculation <i>Gyan Niketan, CBSE</i> , Patna (BR), India CGPA: 10.0/10.0	April, 2012
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